Week 4 Graded assignment Solution
Mathematics for Data Science - 1

1 Multiple Choice Questions (MSQ):

1. Let $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ be two functions, defined as $f(x) = x^3 - 8x^2 + 7$ and $g(x) = -2f(x)$ respectively. Choose the correct option(s) from the following

   - ○ f has two turning points and there are no turning points with negative $y-$coordinate.
   - ○ g has two turning points and $y-$coordinate of only one turning point is positive.
   - ○ g has two turning points and there are no turning points with negative $y-$coordinate.
   - ○ f is strictly increasing in $[10, \infty)$.

$f(x) = x^3 - 8x^2 + 7$

By hit and trial Method: $f(1) = 1 - 8 + 7 = 0$.

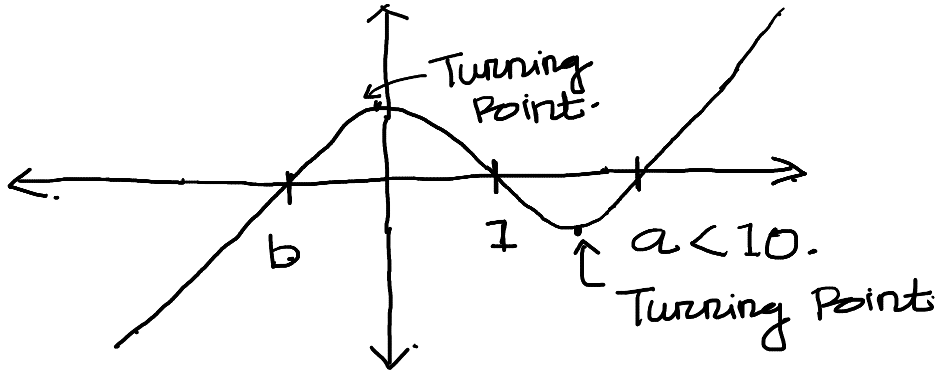
$$x^3 - 8x^2 + 7 = x^3 - x^2 - 7x^2 + 7$$
$$= x^2(x-1) - 7(x^2-1) = (x-1)(x^2 - 7(x+1))$$
$$= (x-1)(x^2 - 7x - 7).$$

$f(x)$ - has three real distinct roots: Say, $1, a, b$.

Note: a, b are less than 10.

If $x \mapsto +\infty \Rightarrow x^3 \mapsto +\infty \Rightarrow f(x) \mapsto +\infty$.

Then the graph of $f(x)$ is similar to;



(Graph of F(x)).

It is clear from the graph that "f" has two turning point.

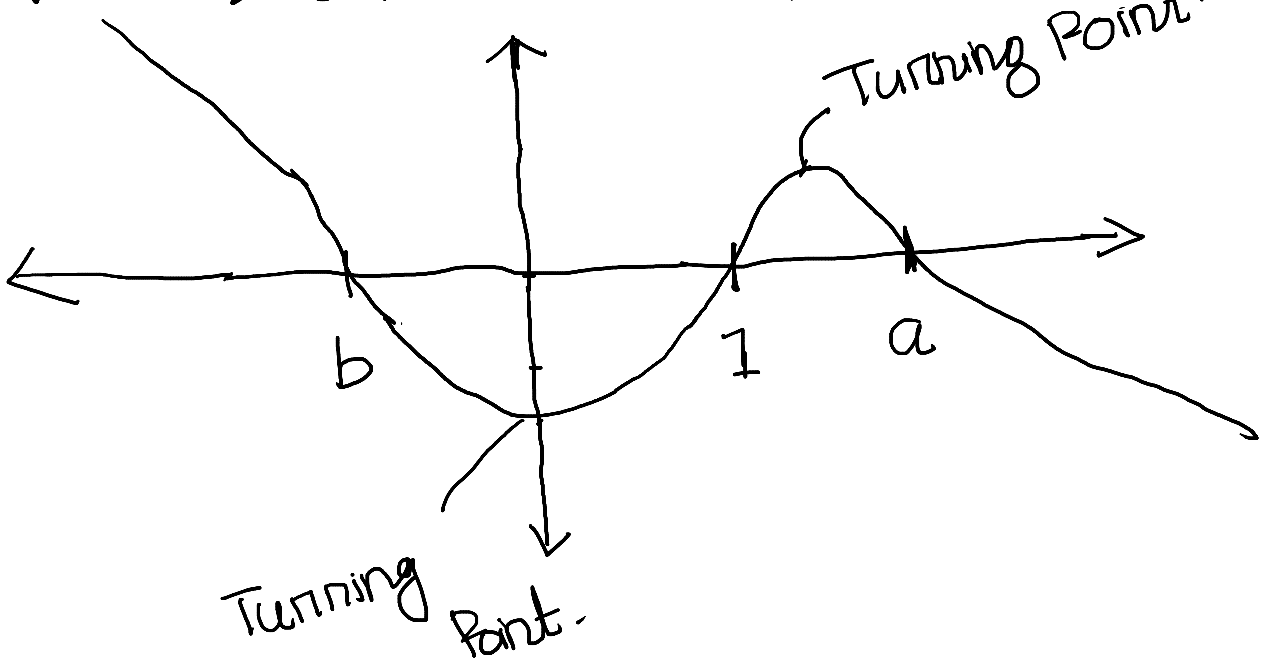
→ Y-coordinate of one of its turning point is +ve.

→ Y-coordinate of one of its turning point is -ve.

• Hence option-1 is incorrect and option-4 is correct.

$$g(x) = -2F(x).$$

The graph of g(x) is similar to;



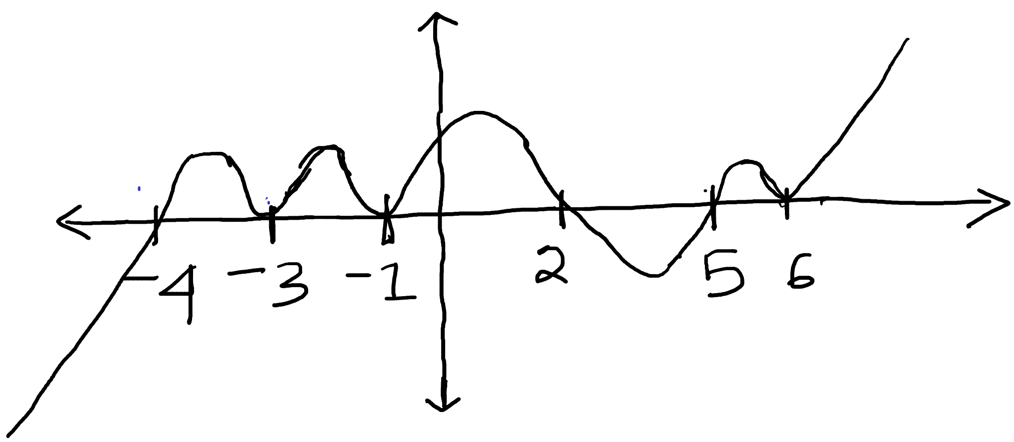
Hence option-2 is correct & option-3 is incorrect

2. Which among the following function first increases and then decreases in all the intervals $(-4,-3)$ and $(-1,2)$ and $(5,6)$.

- ○ $\frac{-1}{10000}(x+1)^2(x-2)(x+3)^2(x+4)(5-x)(x-6)^2$.
- ○ $\frac{1}{10000}(x+1)^2(x-2)(x+3)(x+4)(x-5)^2(x-6)^2(x+7)^2$.
- ○ $\frac{1}{10000}(x+1)^2(x-2)(x+3)^2(x+4)(x-5)^2(x-6)^2$
- ○ $\frac{-1}{10000}(x+1)^2(x-2)(x+3)^2(x+4)^2(5-x)^2(x-6)^2(3-x)$

We will Solve this question using the graph of functions given in each options.

Option 1:



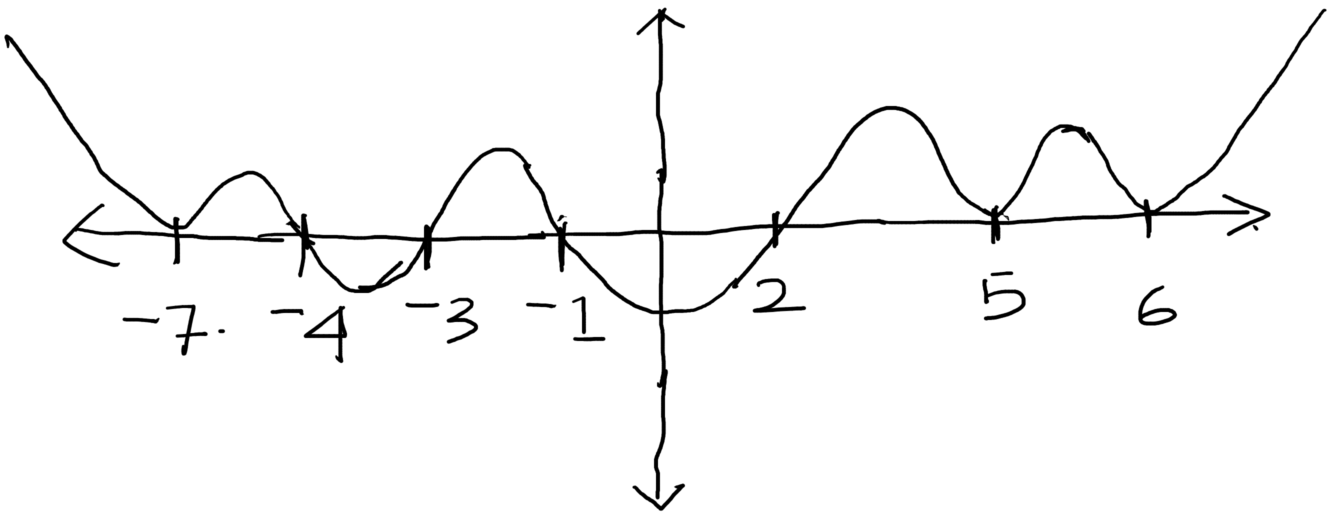
In this Case f(x) increases and then decreases In all the intervals

$(-4,-3) \cup (-1,2) \cup (5,6)$.

Hence option-1 is correct.

Option 2:

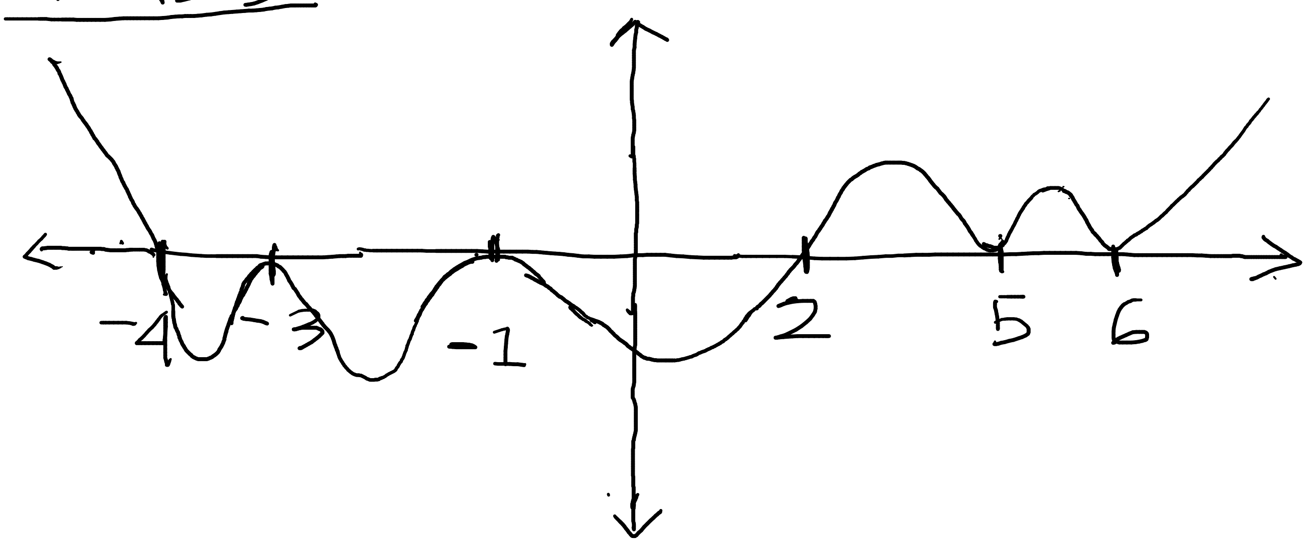
$-1, 2, -3, -4, 5, 6, -7.$



In $(-4,-3)$ & $(-1,2)$, the function first decreases and then increases.

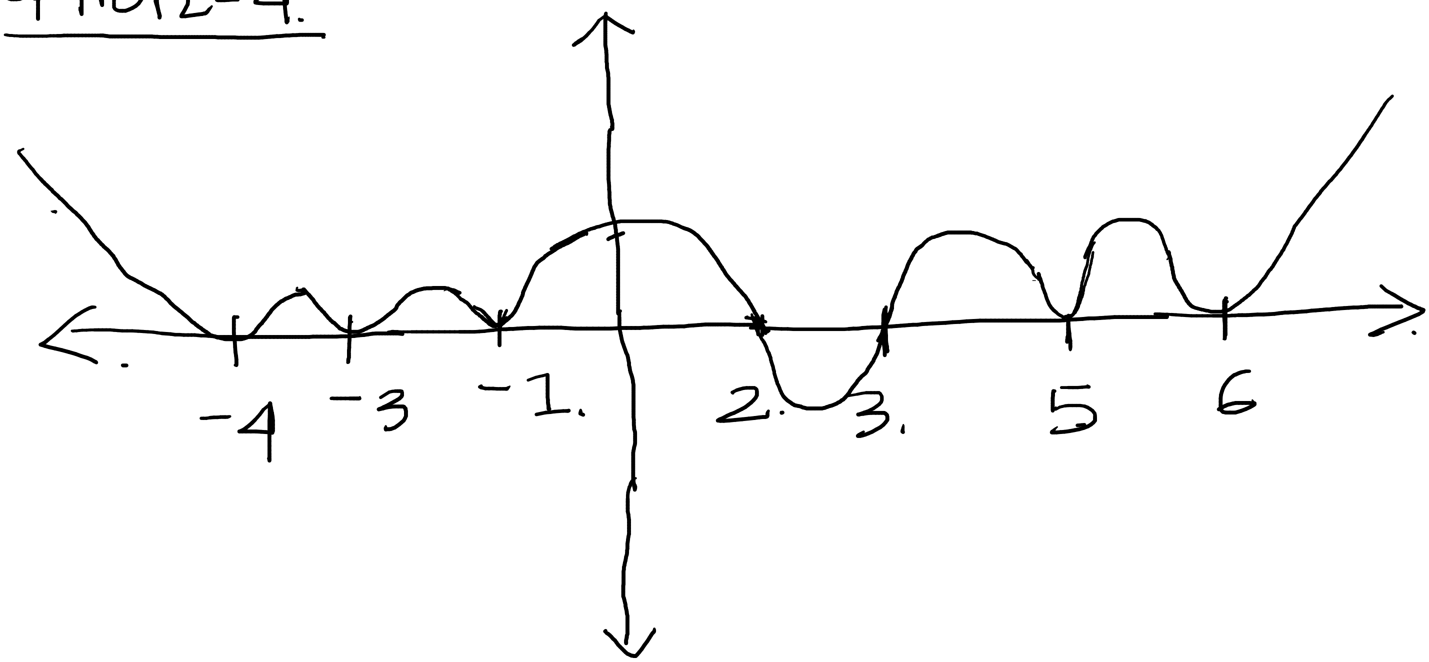
Hence option-2 is incorrect.

Option-3



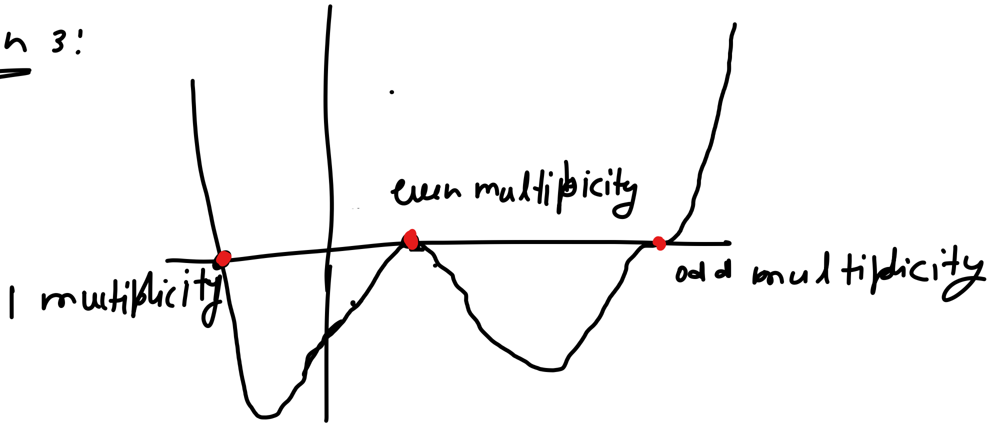
Option-3 is incorrect.

Option-4.



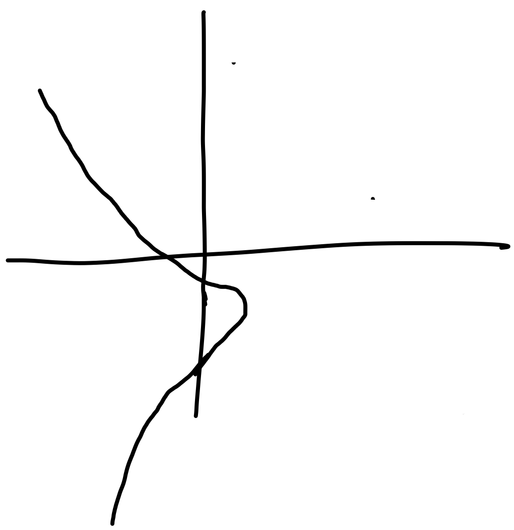
Hence option-4 is correct.

Question 3:



Observe that given graph is a continuous curve. So given graph is of a polynomial with three distinct root. The minimum possible degree could be $1+2+3 = 6$.

Hence have the minimum 6 zero.



Observe that for given single 0 as input and getting two different output.

Hence fig-2 does not represent a function.
Hence fig-2 does not represents a polynomial.

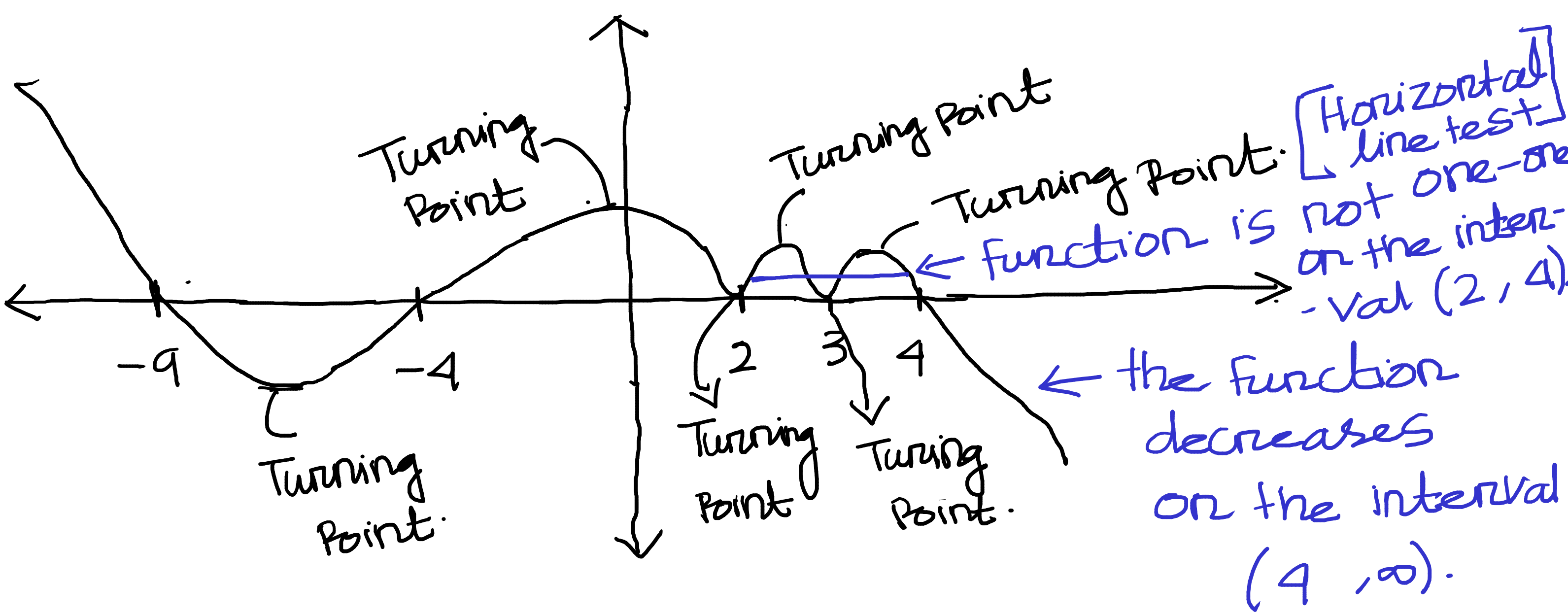
4. Consider a polynomial function $p(x) = -\left(x^2 - 16\right)(x-3)^2(2-x)^2(x+9)$. Choose the set of correct options.

- ○ There are exactly 7 points on $p(x)$ where the slope of the tangent is 0.
- ○ There are exactly 6 points on $p(x)$ where the slope of the tangent is 0.
- ○ $p(x)$ is strictly decreasing when $x \in (4, \infty)$.
- ○ $p(x)$ is one-one function when $x \in (2, 4)$.

- $P(x) = -(x^2-16)(x-3)^2(2-x)^2(x+9)$

$$= -(x+4)(x-4)(x-3)^2(x-2)^2(x+9)$$

The graph of $P(x)$ is similar to ; $\quad 4, -4, 3, 2, -9$



- It is clear from the graph that the function has '6' turning points and there are exactly 6 points where the Slope of the function is zero.

Hence option-2 is correct & option-1 is incorrect.

- Clear from the graph that the function is strictly decreasing on $(4, \infty)$.

  Hence option-3 is correct.

- By Horizontal line test function is not one-one on $(2, 4)$.

  option-4 is incorrect.

Question 5.

$$h(t) = (-0.01t^3 + 0.35t^2 - 3.5t + 10)(t+5)^2 (t-5)(t+1)(2-t)^3$$

$$\Rightarrow h(t) = \frac{1}{100}(-t^3 + 35t^2 - 350t + 1000)(t+5)^2(t-5)(t+1)(2-t)^3$$

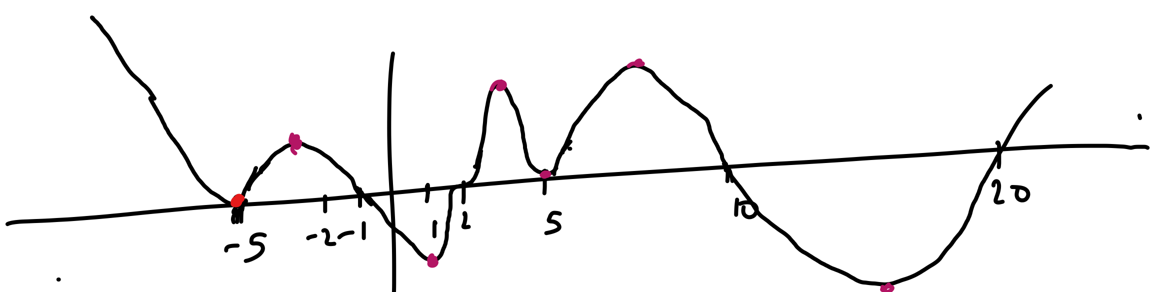
$$= \frac{1}{100}(-1(t^3-1000) + 35t(t-10))(t+5)^2(t-5)(t+1)(2-t)^3$$

$$= \frac{1}{100}(-1(t-10)(t^2+10t+100) + 35t(t-10))(t+5)^2(t-5)(t+1)(2-t)^3$$

$$= \frac{1}{100}(t-10)(t^2-25t+100)(t+5)(t-5)(t+1)(2-t)^3$$

$$= \frac{1}{100}(t-10)(t-20)(t-5)^2(t+5)^2(t+1)(2-t)^3$$

Rough graph would be



Observe that the number of turning point is 7.

Question-6 Observe from the graph

Option 1 The roller coaster will first go up then down in the interval $(-5,-1)$

Similarly for other option we can check from the graph.

7. An ant named B, wants to climb an uneven cliff and reach its anthill (i.e., home of ant). On its way home, B makes sure that it collects some food. A group of ants have reached the food locations which are at $x-$intercepts of the function $f(x) = (x^2-19)\left((x-9)^3-1\right)$. As ants secrete pheromones (a form of signals which other ants can detect and reach the food location), B gets to know the food location. Then the sum of the x-coordinates of all the food locations is

• x-coordinate of all the food locations are same as the roots of the polyno-mial

$$P(x) = (x^2-19)\left((x-9)^3-1\right).$$

$P(x)=0 \Rightarrow x^2-19=0$ or $(x-9)^3-1=0$

$\Rightarrow x = \pm\sqrt{19}$

only real root of $(x-9)^3=1$ is 1.

$\Rightarrow (x-9) = 1.$

$\Rightarrow x = 9+1=10$

Sum of x-coordinates of the food locations

= sum of roots of the polynomial $P(x)$.

$= \sqrt{19} - \sqrt{19} + 10 = 10.$

8. The Ministry of Road Transport and Highways wants to connect three aspirational districts with two roads r_1 and r_2. Two roads are connected if they intersect. The shape of the two roads r_1 and r_2 follows polynomial curve $f(x) = (x-19)(x-17)^2$ and $g(x) = -(x-19)(x-17)$ respectively. What will be the x-coordinate of the third aspirational district, if the first two are at x-intercepts of $f(x)$ and $g(x)$.

- We need find the x-coordinate of the points where the two roads r_1 and r_2 intersect. This is same as the values of x for which

$$f(x) = g(x).$$

$$\Rightarrow (x-19)(x-17)^2 = -(x-19)(x-17).$$

$$\Rightarrow (x-19)(x-17)^2 + (x-19)(x-17) = 0$$

$$\Rightarrow (x-19)(x-17)(x-17+1) = 0$$

$$\Rightarrow (x-19)(x-17)(x-16) = 0$$

$$\Rightarrow x = 19, 17, 16.$$

x-coordinates of the x-intercepts of $f(x)$ & $g(x)$ are 19, 17.

Hence the x-coordinate of the third aspirational district is 16.

Qustion_9

Volume

$V(x) = 4x^3 + 16x^2 + 17x + 5$

Length of the box $L(x) = x+1$

$W(x)$ and $h(x)$ are the width and height of the box.

Observe $V(x) = L(x)\,W(x)\,h(x)$

so $L(x)$ divides $V(x)$

$$
\begin{array}{r|l}
 & 4x^2 + 12x + 5 \\
x+1 & 4x^3 + 16x^2 + 17x + 5 \\
 & \underline{-\cancel{4x^3} \pm 4x^2} \\
 & 12x^2 + 17x + 5 \\
 & \underline{-\cancel{12x^2} \pm 12x} \\
 & 5x + 5 \\
 & \underline{-5x - 5} \\
 & 0
\end{array}
$$

Hence $V(x) = (x+1)(4x^2 + 12x + 5)$

$= (x+1)(4x^2 + 10x + 2x + 5)$

$= (x+1)(2x+5)(2x+1)$

Hence $W(x) = (2x+1)$

$h(x) = 2x+5$

Qustion 10

Area $= L(x) \cdot W(x)$

$= (x+1)(2x+1)$

$= 2x^2 + x + 2x + 1$

$= 2x^2 + 3x + 1$

Qustion 11 $\quad s(t) = t^3 - 2t^2 - 24t + 5000$

To reach 5000 subscribers

$s(t) = 5000$

$\Rightarrow \quad t^3 - 2t^2 - 24t + 5000 = 5000$

$\Rightarrow \quad t(t^2 - 2t - 24) = 0$

$\Rightarrow \quad t(t-6)(x+4) = 0$

$\Rightarrow t = 0, 6, -4$

bc t can be not 0 and negative

Hence after 6 months 5000 subscribers will reach.

Qustion 12 $\quad$ After one year number of subscribers

$s(12) = 12^3 - 2 \times 12^2 - 24 \times 12 + 5000$
$= 6152$

Since polynomial is over estimates by 60%, so let number of actual subscribers is x then

$x + x \times \frac{60}{100} = s(12)$

$\Rightarrow \quad 1.6x = 6152$

$\Rightarrow \quad x = 3845.$

So actual number of subscribers $= 3845.$

Week 4
Graded Assignments Solutions for Maths-1

1 Multiple Select Question (MSQ)

1. Consider the line (L_x) and parabola (P_x) as shown in below figure.

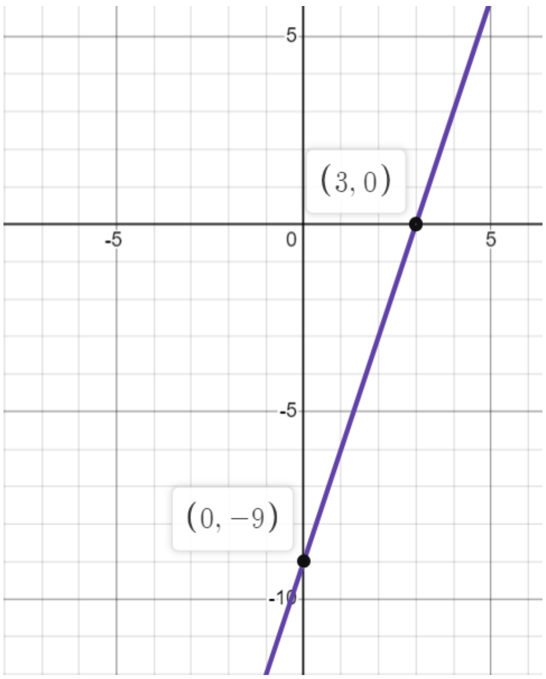


Figure: Line(L_x)

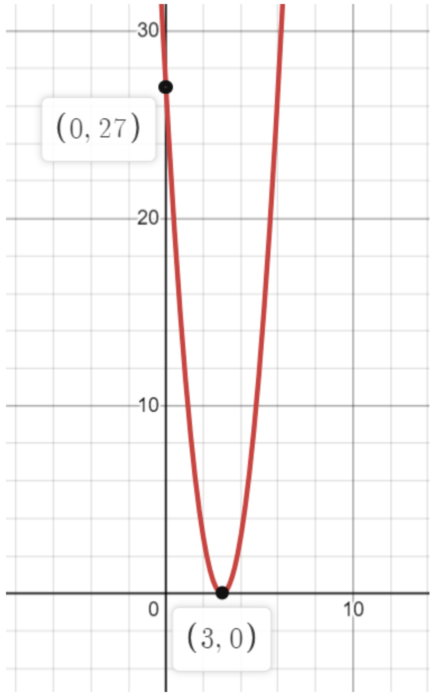
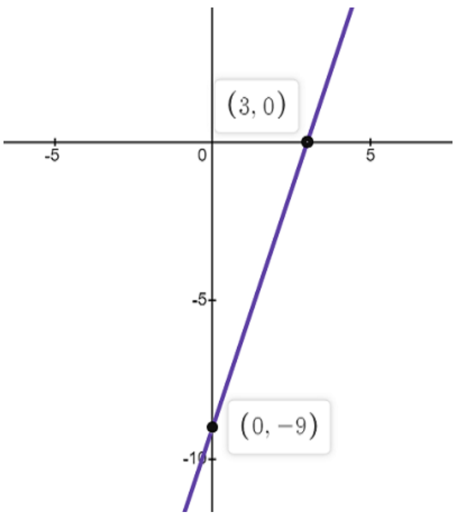
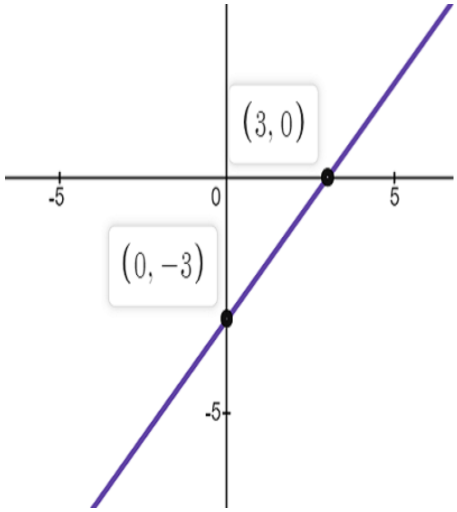


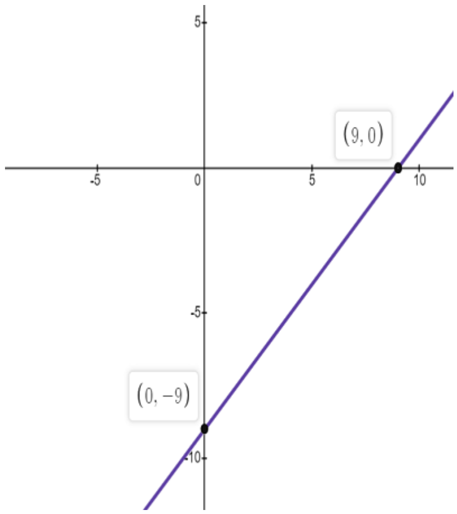
Figure: Parabola (P_x))

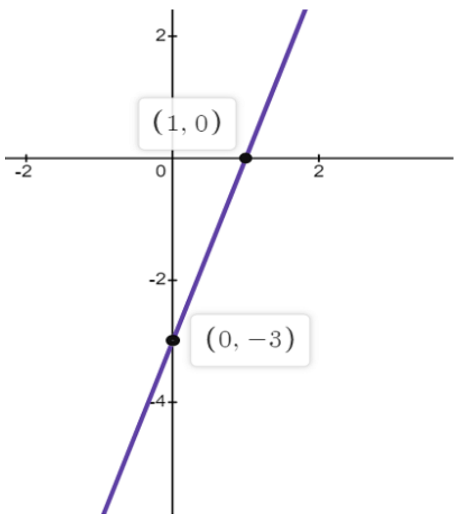
Which among the following represents the graph of $\frac{P_x}{L_x}$?

Answer: Option b [MSQ:1 marks]



○ 

○ 

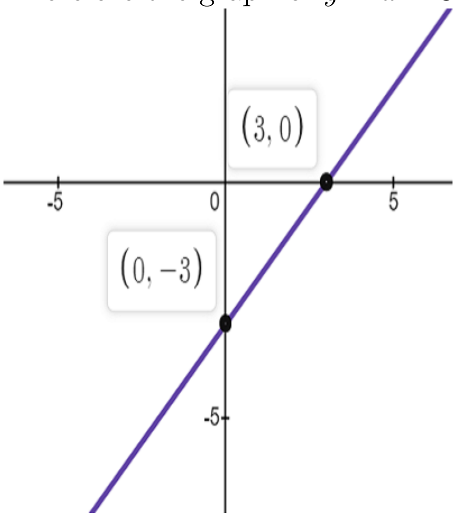
○ 

Solution:
The equation of L_x from the graph is $f(x) = 3x - 9$

The equation of P_x from the graph is $f(x) = 3x^2 - 18x + 27$

$$\frac{P_x}{L_x} = \frac{3x^2 - 18x + 27}{3x - 9} = \frac{x^2 - 6x + 9}{x - 3} = \frac{(x-3)^2}{x-3} = x - 3$$

Therefore the graph of $y = x - 3$ is



Thus, option 2 is correct.

2. Consider $f(x) = x^3 - 4x^2 - 17x + 60$ and $g(x) = x^3 + 5x^2 - 8x - 12$, whose one of the roots are given in the set $\{3, 2, -3, -2\}$. Choose the set of correct options regarding $f(x)$ and $g(x)$.

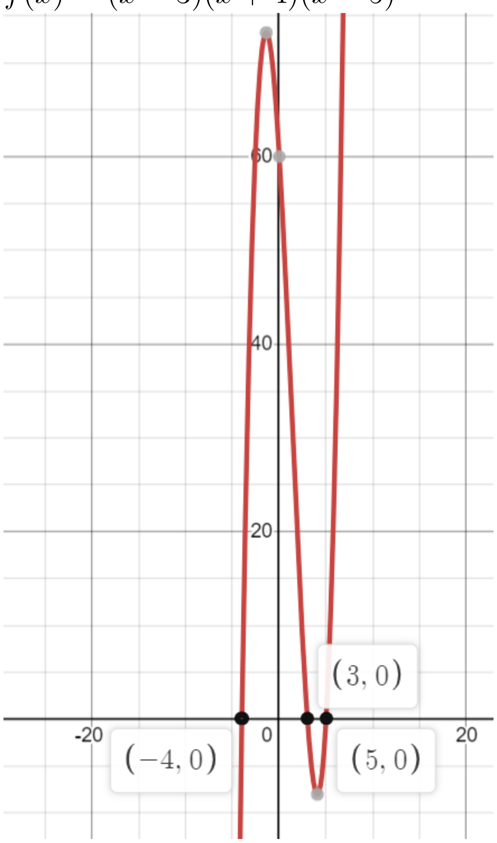
Answer: Option b [MSQ:1 marks]

- ◯ The $f(x)$ cuts the X-axis at 3, 5 and 4.
- ◯ The $g(x)$ cuts the X-axis at 2, -6 and -1.
- ◯ If $x \in (-6, 2)$, then $g(x)$ is positive.
- ◯ $f(x)$ is negative when $x \in [-4, 3] \cup (3, \infty)$.
- ◯ $g(x)$ is positive when $x \in (-\infty, -6) \cup (-1, -2)$.

Solution: From the given set, $x = 3$ is one of the root of $f(x)$, since $f(3) = 0$. To find the other roots: $\dfrac{x^3 - 4x^2 - 17x + 60}{x - 3} = x^2 - x - 20 \Rightarrow x = -4, 5$ So, option a is incorrect.

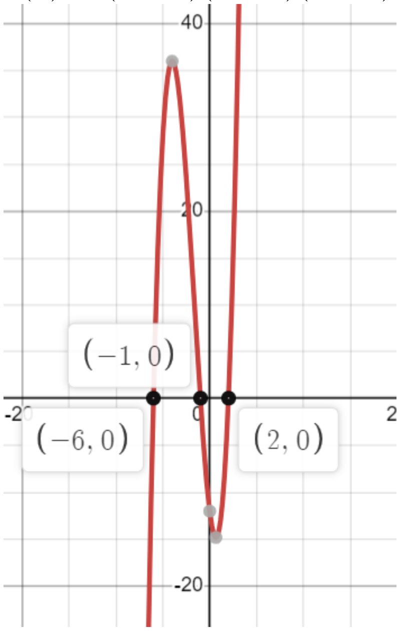
From the given set, $x = 2$ is one of the root of $g(x)$, since $g(2) = 0$. To find the other roots: $\dfrac{x^3 + 5x^2 - 8x - 12}{x - 2} = x^2 + 7x + 6 \Rightarrow x = -6, -1$ So, **option b is correct**.

$f(x) = (x - 3)(x + 4)(x - 5)$



$f(x)$ is positive when $x \in (-4, 3) \cup (5, \infty)$.

Thus, option d is incorrect.
$g(x) = (x-2)(x+6)(x+1)$



$g(x)$ is negative when $x \in (-\infty, -6) \cup (-1, -2)$.
So, option c and e are incorrect.

3. Consider a rectangular box whose length is x, height is 3 units less than length, and breadth is 5 units more than height. If the volume of the rectangular box is 24 $unit^3$, then choose the correct value of x from the given options.

Answer: Option b [NAT:1 marks]

- ○ 3
- ○ 4
- ○ 5
- ○ 2

Solution:

Let the length of rectangular box be x, then the height is $h = x - 3$
and breadth is $b = h + 5 = x - 3 + 5 = x + 2$. The volume of rectangular box is given by $l \times b \times h$.
$V = x(x-3)(x+2) = x(x^2) = x(x^2 - x - 6) = x^3 - x^2 - 6x$.
Given that volume of the rectangular box is 24 $unit^3$, that is,
$x^3 - x^2 - 6x = 24$

So, from the given options, option b satisfied the given equation.

4. Consider a polynomial function $f(x)$ of degree 4 which intersects the X-axis at $x = 2, x = -3$ and $x = -4$. Moreover, $f(x) < 0$ when $x \in (1, 2)$, and $f(x) > 0$ when $x \in (-1, 1)$. Find out the equation of the polynomial.

   Answer: Option b [MSQ:1marks]

   - ◯ $a(x-2)^2(x^2+7x+12), a > 0$
   - ◯ $a(x^4+4x^3-7x^2-22x+24), a > 0$
   - ◯ $a(x-2)^2(x^2+2x-8), a > 0$
   - ◯ $a(x^4-5x^3-7x^2-50x-24), a > 0$

   Solution:

   Given that $f(x)$ has degree 4 which intersects the X-axis at $x = 2, x = -3$ and $x = -4$. And also $f(x) < 0$ when $x \in (1, 2)$, and $f(x) > 0$ when $x \in (-1, 1)$. It means that when $x = 1$, $f(x) = 0$..

   Hence, the equation of $f(x)$ is given by

   $f(x) = a(x-2)(x+3)(x+4)(x-1)$, where $a > 0$

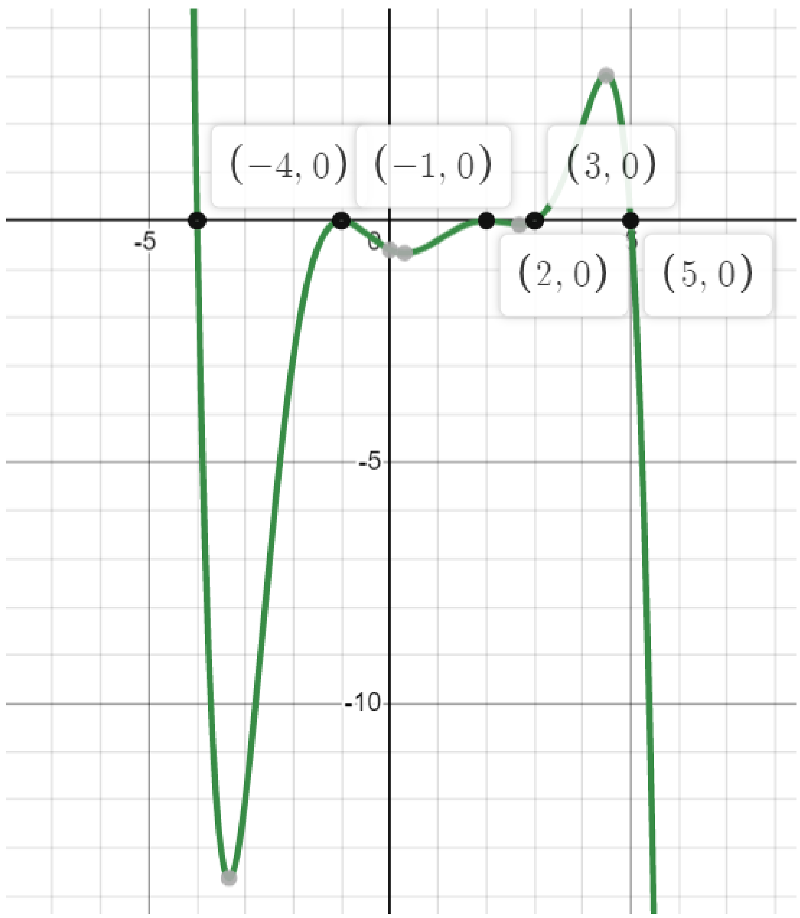
   $= a(x^4+4x^3-7x^2-22x+24)$.

   Hence , option b is correct.

5. Consider a polynomial function $f(x) = \frac{-1}{300}(x-2)^2(x-3)(x+1)^2(x+4)(x-5)$. Choose the correct set of options.
Answer: Option b,c,d and e [MSQ:1 marks]

- ◯ The function $f(x)$ has exactly 7 turning points.
- ◯ The function $f(x)$ has exactly 6 points where the slope is 0.
- ◯ The function $f(x)$ is neither even nor odd function.
- ◯ In the interval $x \in (3, 5)$, $f(x)$ is increasing first and then decreasing.
- ◯ The function $f(x)$ is negative when $x \in (-1, 2)$.

Solution:



From the above graph,

- We can seen that $f(x)$ has exactly 6 turning points. So, option b is correct.
- In the interval $x \in (3, 5)$, $f(x)$ is increasing first and then decreasing. So, option d is correct.

- The function $f(x)$ is negative when $x \in (-1, 2)$. So, option e is correct.

Also, if we replace x by $-x$ in $f(x)$, then $f(-x) \neq -f(x)$ or $f(-x) \neq f(x)$. So, $f(x)$ is neither even nor odd function.

6. Consider a polynomial function $P(x) = (x^4 + 4x^3 + x + 10)$ and $Q(x) = (x^3 + 2x^2 - 6)$. If $M(x)$ is the equation of the straight line passing through $(2, Q(2))$ and having slope 3, then find out the equation of $P(x) + M(x)Q(x)$.
   Choose the correct option.
   Answer: Option A [MSQ:1 marks]

   - ◯ $4x^4 + 14x^3 + 8x^2 - 17x - 14$
   - ◯ $4x^4 + 14x^3 - 6x^2 - 19x - 34$
   - ◯ $4x^4 + 2x^3 + 8x^2 - 17x - 14$
   - ◯ $4x^4 + 2x^3 + 8x^2 - 18x - 34$

   Solution:
   $M(x)$ is the equation of the straight line passing through $(2, Q(2)) = (2, 10)$ and having slope 3.
   The equation of M will be , $M(x) =$ 3x +4
   $P(x) + M(x)Q(x) = (x^4 + 4x^3 + x + 10) + (3x + 4)(x^3 + 2x^2 - 6)$
   $= (x^4 + 4x^3 + x + 10) + (3x^4 + 6x^3 - 18x + 4x^3 + 8x^2 - 24)$
   $= 4x^4 + 14x^3 + 8x^2 - 17x - 14$

   Thus, option A is correct.

2 Numerical Answer Type (NAT):

7. A function $f(x)$ which is the best fit for the data given in the Table-1 recorded by a student, is

$$f(x) = -\left(x-1\right)^2\left(x-3\right)\left(x-5\right)\left(x-7\right)+c$$

What will be the value of c, so that SSE (Sum Squared Error) will be minimum?
(NAT)(**Ans: 3.4**) [Marks:1]

x	1	2	3	4	5
y	4	18	4	-24	3

Table-1

Solution:

x_i	y_i	$f(x_i)$	$(y_i - f(x_i))^2$
1	4	c	$(4-c)^2$
2	18	$15+c$	$(3-c)^2$
3	4	c	$(4-c)^2$
4	-24	$-27+c$	$(3-c)^2$
5	3	c	$(3-c)^2$

The SSE (Sum Squared Error) is given by
$\sum_{i=1}^{5}(y_i - f(x_i))^2 = 3(3-c)^2 + 2(4-c)^2$
$=3(9-6c+c^2)+2(16-8c+c^2) = 27-18c+3c^2+32-16c+2c^2$
$= 5c^2 - 34c + 59$
This is a quadratic equation and it will have minimum value at the vertex.
Therefore, SSE will be minimum at, c $= \dfrac{-b}{2a} = \dfrac{34}{10} = 3.4$

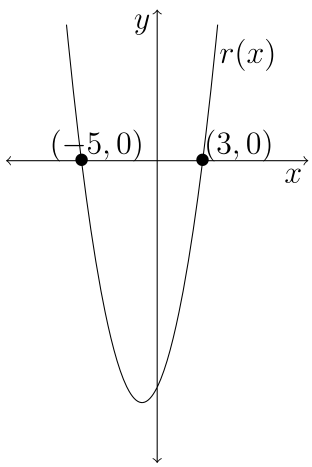
8. An ant named B, wants to climb an uneven cliff and reach its anthill (i.e., home of ant). On its way home, B makes sure that it collects some food. A group of ants have reached the food locations which are at $x-$intercepts of the function $f(x) = (x^2 - 19)\left((x-9)^3 - 1\right)$. As ants secrete pheromones (a form of signals which other ants can detect and reach the food location), B gets to know the food location. Then the sum of the x-coordinates of all the food locations is
(NAT) (Answer: 10)

Solution:
Given that, a group of ants have reached the food locations which are at $x-$intercepts, which means, x-coordinates of all the food locations are same as the roots of the polynomial, $f(x) = (x^2 - 19)\left((x-9)^3 - 1\right)$
$(x^2 - 19)\left((x-9)^3 - 1\right) = 0$
$(x^2 - 19) = 0$ or $((x-9)^3 - 1) = 0$
$(x^2 - 19) = 0 \Rightarrow x = \sqrt{19}, -\sqrt{19}$
$((x-9)^3 - 1) = 0 \Rightarrow x = 10$

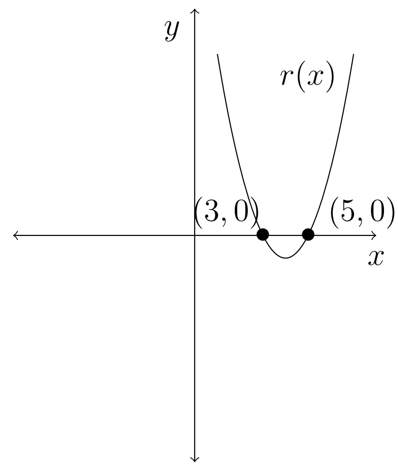
The sum of the x-coordinates of all the food locations
= sum of roots of polynomials $f(x) = \sqrt{19} - \sqrt{19} + 10 = 10$

9. The Ministry of Road Transport and Highways wants to connect three aspirational districts with two roads r_1 and r_2. Two roads are connected if they intersect. The shape of the two roads r_1 and r_2 follows polynomial curve $f(x) = (x-19)(x-17)^2$ and $g(x) = -(x-19)(x-17)$ respectively. What will be the $x-$coordinate of the third aspirational district, if the first two are at $x-$intercepts of $f(x)$ and $g(x)$.
(NAT) [Ans: 16]
Solution:
We need to find the x-coordinate of the points where the two roads r_1 and r_2 intersect. This is same as same as the values of x for which,
$f(x) = g(x)$
$(x-19)(x-17)^2 = -(x-19)(x-17)$
$(x-19)(x-17)^2 + (x-19)(x-17) = 0$
$(x-19)(x-17)(x-17+1) = 0$
$(x-19)(x-17)(x-16) = 0$
$x = 19, 17, 16$
Hence, the x-coordinate of the third aspirational district is 16.

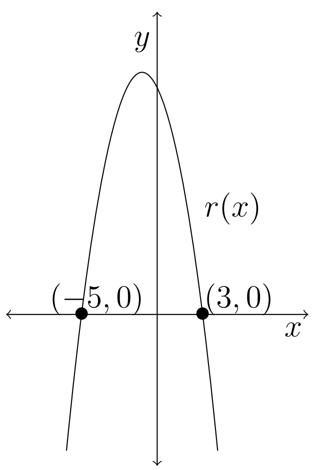
10. Let $r(x)$ be a polynomial function which is obtained as the quotient after dividing the polynomial $p(x) = (x+5)(x-3)(x^2-4)$ by the polynomial $q(x) = (x-2)(2+x)$. Choose the correct option which represents the polynomial $r(x)$ most appropriately. (MSQ)(Ans: Option 1) [Marks:1]



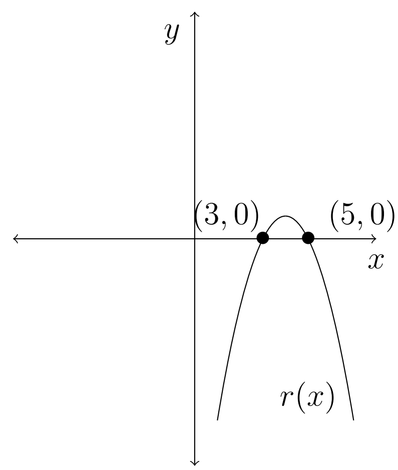
Option 1



Option 2



Option 3

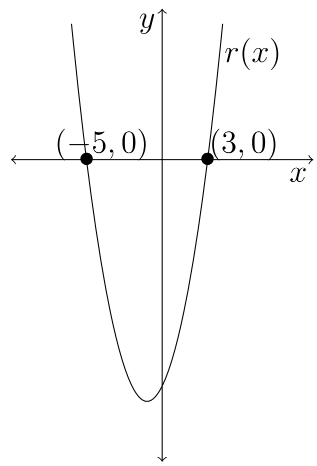


Option 4

Solution: $p(x) = (x+5)(x-3)(x^2-4)$ and $q(x) = (x-2)(2+x)$

$$\frac{p(x)}{q(x)} = \frac{(x+5)(x-3)(x^2-4)}{(x-2)(2+x)} = \frac{(x+5)(x-3)(x+2)(x-2)}{(x-2)(2+x)} = (x+5)(x-3)$$

Therefore, $r(x) = (x+5)(x-3)$. The graph of r(x) is



Option 1 is correct.

Week 4 Graded assignment Solution
Mathematics for Data Science - 1

1 Multiple Choice Questions (MSQ):

1. Let $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ be two functions, defined as $f(x) = x^3 - 8x^2 + 7$ and $g(x) = -2f(x)$ respectively. Choose the correct option(s) from the following

   - ○ f has two turning points and there are no turning points with negative $y-$coordinate.
   - ○ g has two turning points and $y-$coordinate of only one turning point is positive.
   - ○ g has two turning points and there are no turning points with negative $y-$coordinate.
   - ○ f is strictly increasing in $[10, \infty)$.

$f(x) = x^3 - 8x^2 + 7$

By hit and trial Method: $f(1) = 1 - 8 + 7 = 0$.

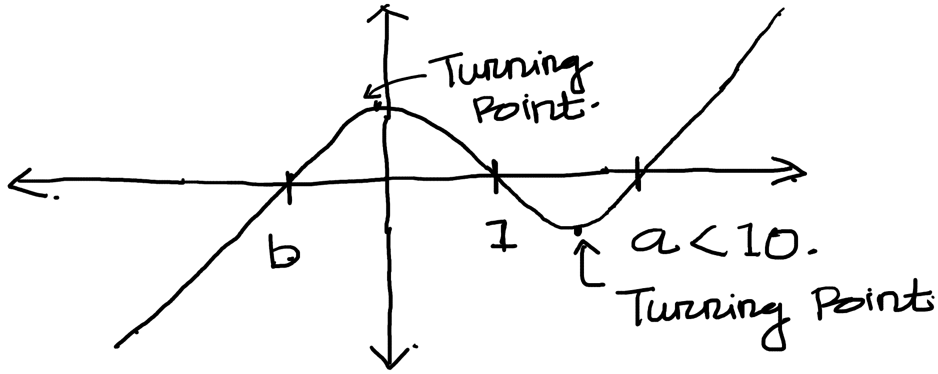
$$x^3 - 8x^2 + 7 = x^3 - x^2 - 7x^2 + 7$$
$$= x^2(x-1) - 7(x^2-1) = (x-1)(x^2 - 7(x+1))$$
$$= (x-1)(x^2-7x-7).$$

$f(x)$ – has three real distinct roots: Say, $1, a, b$.

Note: a, b are less than 10.

If $x \mapsto +\infty \Rightarrow x^3 \mapsto +\infty \Rightarrow f(x) \mapsto +\infty$.

Then the graph of $f(x)$ is similar to;



(Graph of F(x)).

It is clear from the graph that "f" has two turning point.

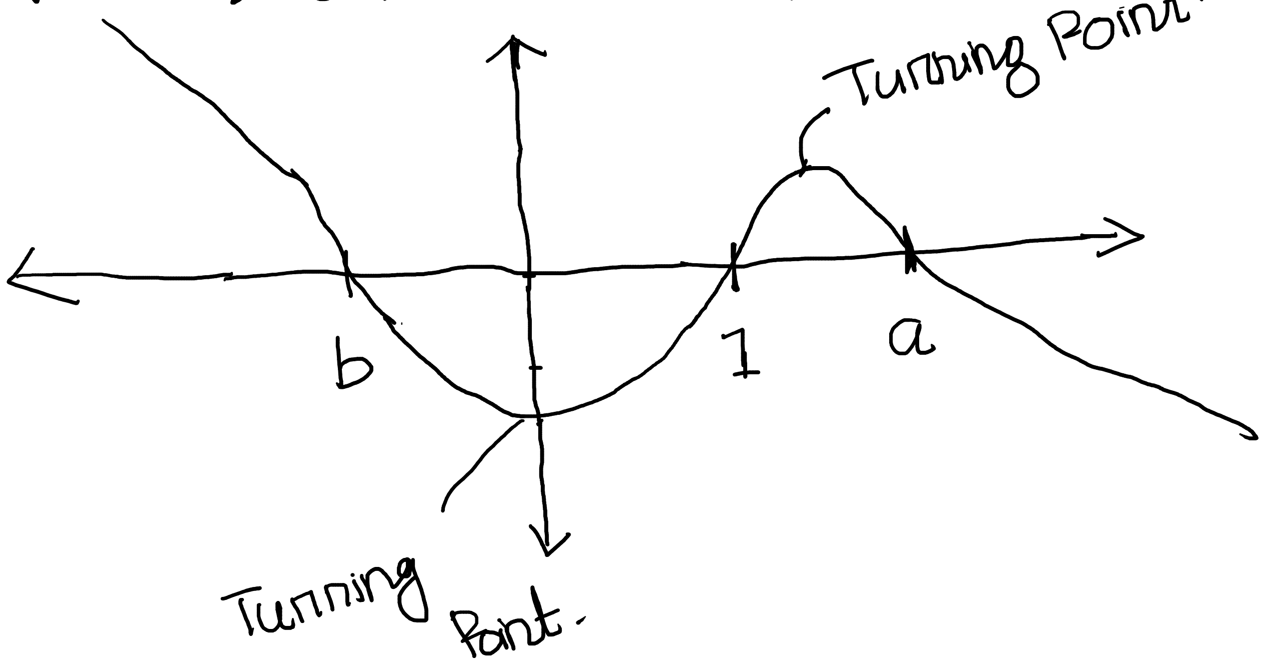
→ Y-coordinate of one of its turning point is +ve.

→ Y-coordinate of one of its turning point is -ve.

• Hence option-1 is incorrect and option-4 is correct.

$$g(x) = -2F(x).$$

The graph of g(x) is similar to;



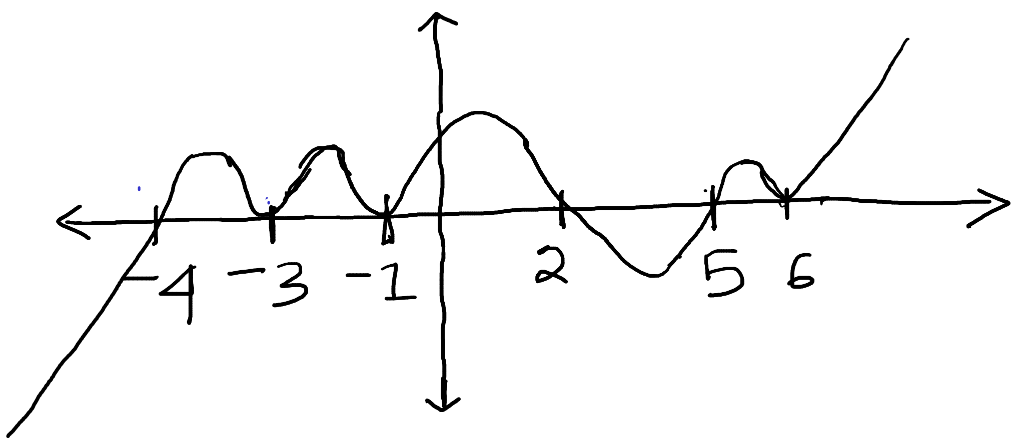
Hence option-2 is correct & option-3 is incorrect

2. Which among the following function first increases and then decreases in all the intervals $(-4,-3)$ and $(-1,2)$ and $(5,6)$.

- ○ $\frac{-1}{10000}(x+1)^2(x-2)(x+3)^2(x+4)(5-x)(x-6)^2$.
- ○ $\frac{1}{10000}(x+1)^2(x-2)(x+3)(x+4)(x-5)^2(x-6)^2(x+7)^2$.
- ○ $\frac{1}{10000}(x+1)^2(x-2)(x+3)^2(x+4)(x-5)^2(x-6)^2$
- ○ $\frac{-1}{10000}(x+1)^2(x-2)(x+3)^2(x+4)^2(5-x)^2(x-6)^2(3-x)$

We will solve this question using the graph of functions given in each options.

Option 1:



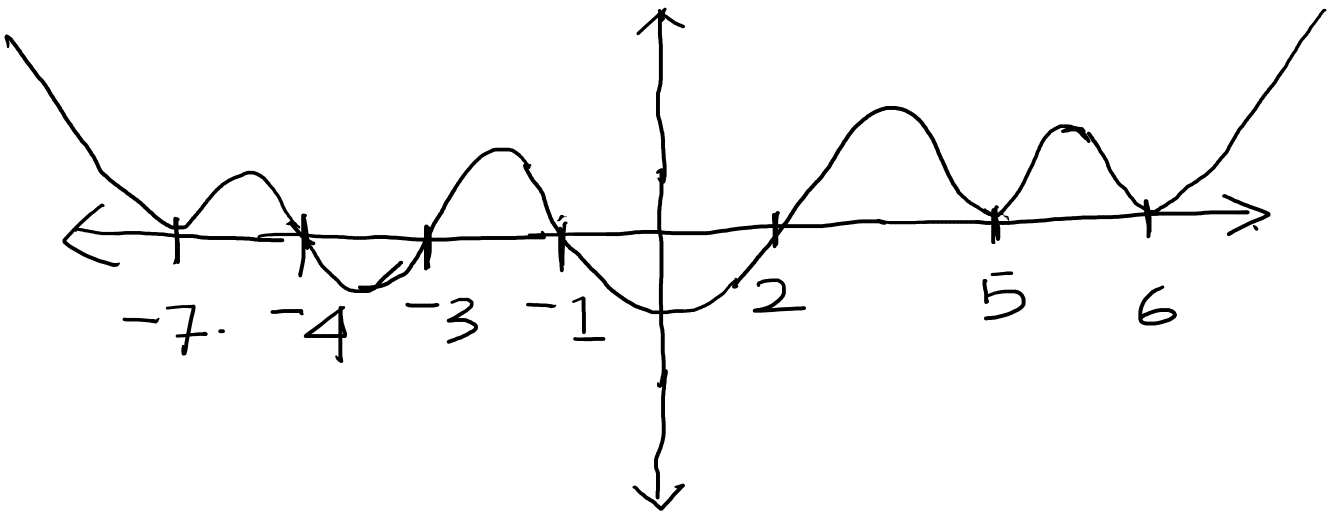
In this case $f(x)$ increases and then decreases in all the intervals

$(-4,-3) \cup (-1,2) \cup (5,6)$.

Hence option-1 is correct.

Option 2:

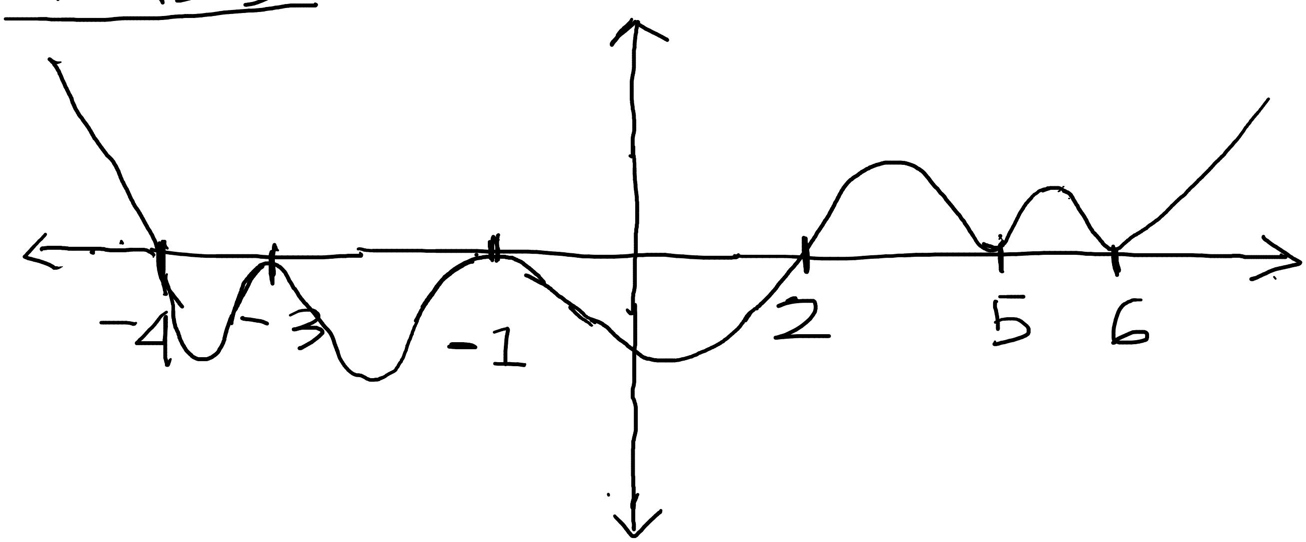
$-1, 2, -3, -4, 5, 6, -7.$



In $(-4,-3)$ & $(-1,2)$, the function first decreases and then increases.

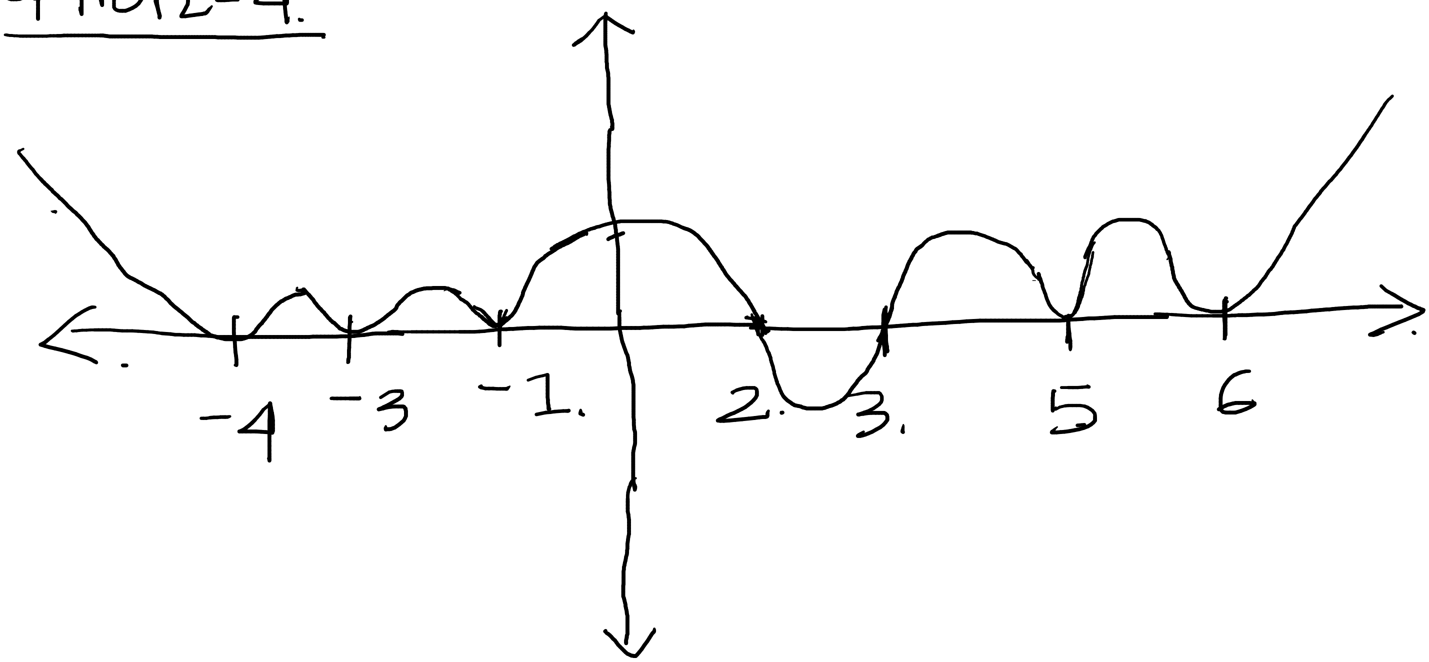
Hence option-2 is incorrect.

Option-3



Option-3 is incorrect.

Option-4.



Hence option-4 is correct.

Question 3

Soln

line $l(x) = x$

$f(x) = x^4 - 13x - 42$

$g(x) = x^2 - x - 6$

lets find $r(x)$ which is quotient polynomial when $f(x)$ is divided by $g(x)$

$$
\begin{array}{r|l}
 & x^2 + x + 8 \\
\hline
x^2 - x - 6 & \cancel{x^4} - 13x - 42 \\
 & \cancel{x^4} \mp x^3 \mp 6x^2 \\
\hline
 & \cancel{x^3} + 6x^2 - 13x - 42 \\
 & \cancel{x^3} \mp x^2 \mp 6x \\
\hline
 & \cancel{8x^2} - 7x - 42 \\
 & \cancel{8x^2} \mp 8x \mp 48 \\
\hline
 & x + 6
\end{array}
$$

So $r(x) = x^2 + x + 8$

Now two find the intersection point

$$l(x) = r(x)$$

$$\Rightarrow \quad x^2 + x + 8 = x$$

$$\Rightarrow \quad x^2 + 8 = 0$$

Observe that this equation has

no root.

So, there is no itersection point.

4. Consider a polynomial function $p(x) = -\left(x^2-16\right)(x-3)^2(2-x)^2(x+9)$. Choose the set of correct options.

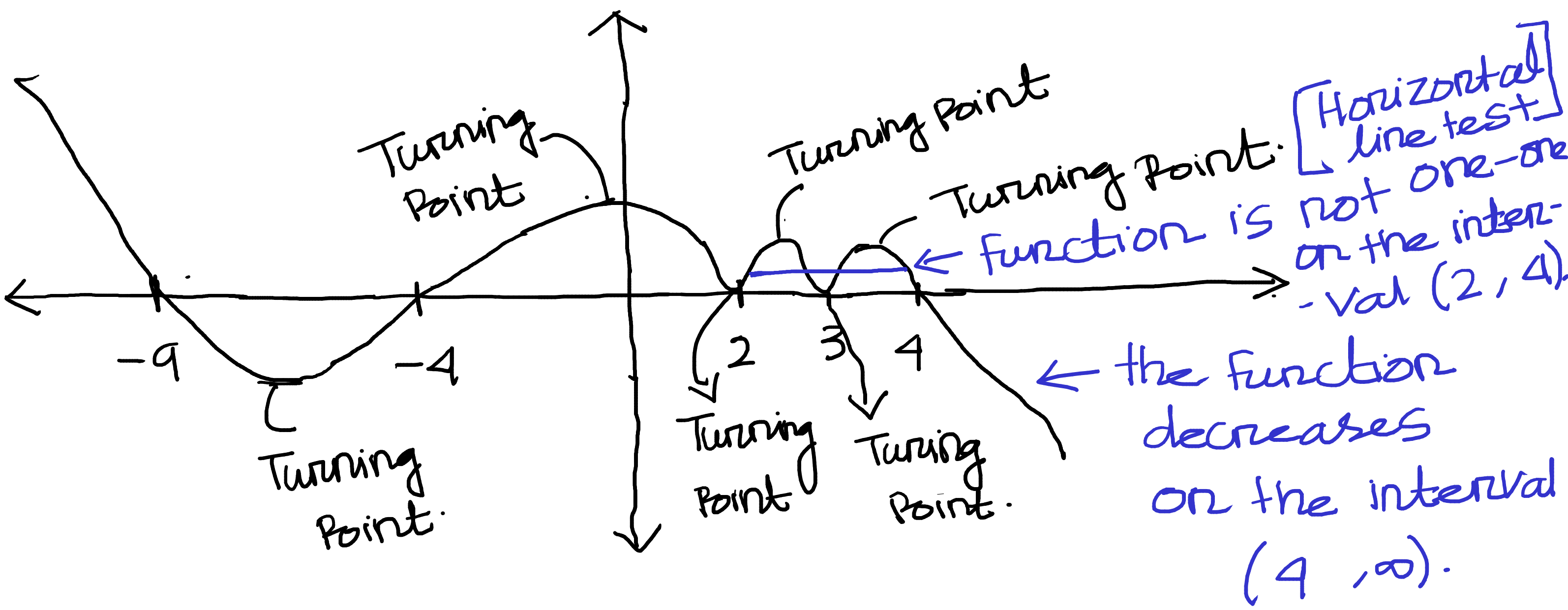
- ○ There are exactly 7 points on $p(x)$ where the slope of the tangent is 0.
- ○ There are exactly 6 points on $p(x)$ where the slope of the tangent is 0.
- ○ $p(x)$ is strictly decreasing when $x \in (4, \infty)$.
- ○ $p(x)$ is one-one function when $x \in (2, 4)$.

- $P(x) = -(x^2-16)(x-3)^2(2-x)^2(x+9)$

$$= -(x+4)(x-4)(x-3)^2(x-2)^2(x+9)$$

4, −4, 3, 2, −9

The graph of P(x) is similar to ;



- It is clear from the graph that the function has '6' turning points and there are exactly 6 points where the slope of the function is zero.

Hence option-2 is correct & option-1 is incorrect.

- Clear from the graph that the function is strictly decreasing on $(4, \infty)$.

Hence option-3 is correct.

- By Horizontal line test function is not one-one on $(2, 4)$.

option-4 is incorrect.

Question 5.

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$$

has following prop:-

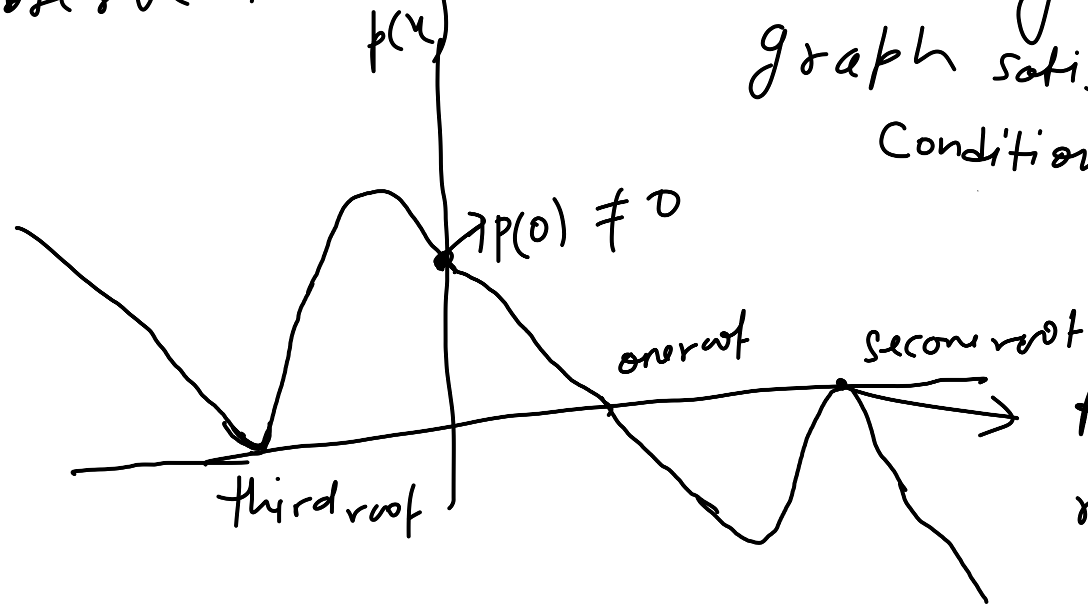
(I) $p(x)$ is an odd degree polynomial with at least three distinct roots.

(II) $p(x)$ has exactly two distinct positive real roots.

(III) $(x-5)^2$ is a factor of $p(x)$

(IV) $p(0) \neq 0$

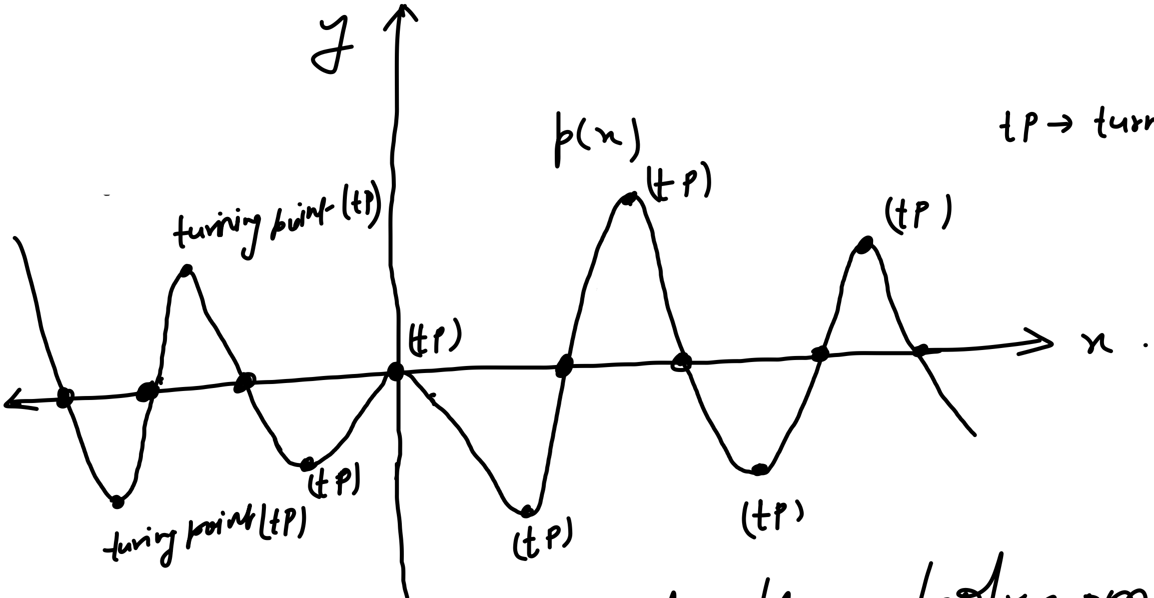
Observe that $p(x)$ is only in this graph satisfies all Conditions.



As 5 is a root with even multiplicity

Question 6

Sol^n



Option 1

Observe from the graph of the polynomial Except at origin rest roots have only one multiplicity as graph of the polynomial is linear when it is intersecting the x-axis. And at the origin graph touches the and bounce back so it 0 is a root with even multiplicity (at least 2 multiplicity).

So p(x) has 8 roots and as 0 is a root with even multiplicity (at least 2 multiplicity) and so degree of p(x) is atleast 9. (Number of roots decides that, that number of factors are there in $p(x)$)).

Option 2 As we calculated above degree of $p(x)$ is atleast 9. In this $p(x)$ is an odd degree polynomial.

Suppose $p(x)$ is an even degree polynomial, In this case surly any one of roots other than zero is also have even multiplicity But except zero root there is no root which have even multiplicity. Hence $p(x)$ is an odd degree polynomial.

Option 3 Turning point are nothing but points where function value is changing from increasing to decreasing or from decreasing to increasing. And from graph it is clear that it have 8 turning points.

Option 4: Observe that multiplicity of root zero is even (as we discussed in option 1) and none of the root has this even multiplicity

7. An ant named B, wants to climb an uneven cliff and reach its anthill (i.e., home of ant). On its way home, B makes sure that it collects some food. A group of ants have reached the food locations which are at $x-$intercepts of the function $f(x) = (x^2 - 19)\left((x-9)^3 - 1\right)$. As ants secrete pheromones (a form of signals which other ants can detect and reach the food location), B gets to know the food location. Then the sum of the x-coordinates of all the food locations is

- x-coordinate of all the food locations are same as the roots of the polyno-
-mial

$$P(x) = (x^2 - 19)\left((x-9)^3 - 1\right).$$

$P(x) = 0 \Rightarrow x^2 - 19 = 0$ or $(x-9)^3 - 1 = 0$

$\Rightarrow x = \pm\sqrt{19}$

only real root of $(x-9)^3 = 1$ is 1.

$\Rightarrow (x-9) = 1.$

$\Rightarrow x = 9 + 1 = 10$

Sum of x-coordinates of the food locations

= Sum of roots of the polynomial $P(x)$.

$= \sqrt{19} - \sqrt{19} + 10 = 10.$

8. The Ministry of Road Transport and Highways wants to connect three aspirational districts with two roads r_1 and r_2. Two roads are connected if they intersect. The shape of the two roads r_1 and r_2 follows polynomial curve $f(x) = (x-19)(x-17)^2$ and $g(x) = -(x-19)(x-17)$ respectively. What will be the $x-$coordinate of the third aspirational district, if the first two are at $x-$intercepts of $f(x)$ and $g(x)$.

- We need find the x-coordinate of the points where the two roads r_1 and r_2 intersect. This is same as the values of x for which

$$f(x) = g(x).$$

$$\Rightarrow (x-19)(x-17)^2 = -(x-19)(x-17).$$

$$\Rightarrow (x-19)(x-17)^2 + (x-19)(x-17) = 0$$

$$\Rightarrow (x-19)(x-17)(x-17+1) = 0$$

$$\Rightarrow (x-19)(x-17)(x-16) = 0$$

$$\Rightarrow x = 19, 17, 16.$$

x-coordinates of the x-intercepts of $f(x)$ & $g(x)$ are 19, 17.

Hence the x-coordinate of the third aspirational district is 16.

Question 9 $\quad M(n) = -\left(\frac{n^2}{1000}\right)(n^3-15n^2+50n)+40$, $n \in \{1,2 \cdots 12\}$

To get number of times Ritwik score 40

$M(n) = 40$

$\Rightarrow \quad -\left(\frac{n^2}{1000}\right)(n^3-15n^2+50n)+40 = 40$

$\Rightarrow \quad n^3-15n^2+50n = 0$

$\Rightarrow \quad n(n^2-15n+50) = 0$

$\Rightarrow \quad n(n-10)(n-5) = 0$

$\Rightarrow \quad n=0$ or $n=10$ or $n=5$

but $n \neq 0$ as $n \in \{1,2, \cdots 12\}$

So two times Ritwik got excatly 40 marks.

Question 10 $\quad$ To get pass Ritwik marks should be greater than 40 i.e

$M(n) \geq 40$.

$\Rightarrow \quad \left(-\frac{n^2}{1000}\right)(n^3-15n^2+50n)+40 \geq 40$

$\Rightarrow \left(\frac{-n^2}{1000}\right)(n^3-15n^2+50n) \geq 0$

$\Rightarrow \quad n^3(n-10)(n-5) \leq 0$

from inequality, we can conclude that if

$n \in \{5,6,7,8,9,10\}$ then $n^3(n-10)(n-5) \leq 0$

So total in 6 mocks test Ritwik passed.